

Not all breakpoints are created equally: Test-retest reliability and bilateral differences of SmO₂ breakpoints during continuous cycling step tests

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1 – Background

NIRS-derived muscle oxygen saturation (SmO₂) shows characteristic breakpoints (BPs) during a graded exercise test (GXT), reflecting shifts in the O₂ delivery-utilization balance.

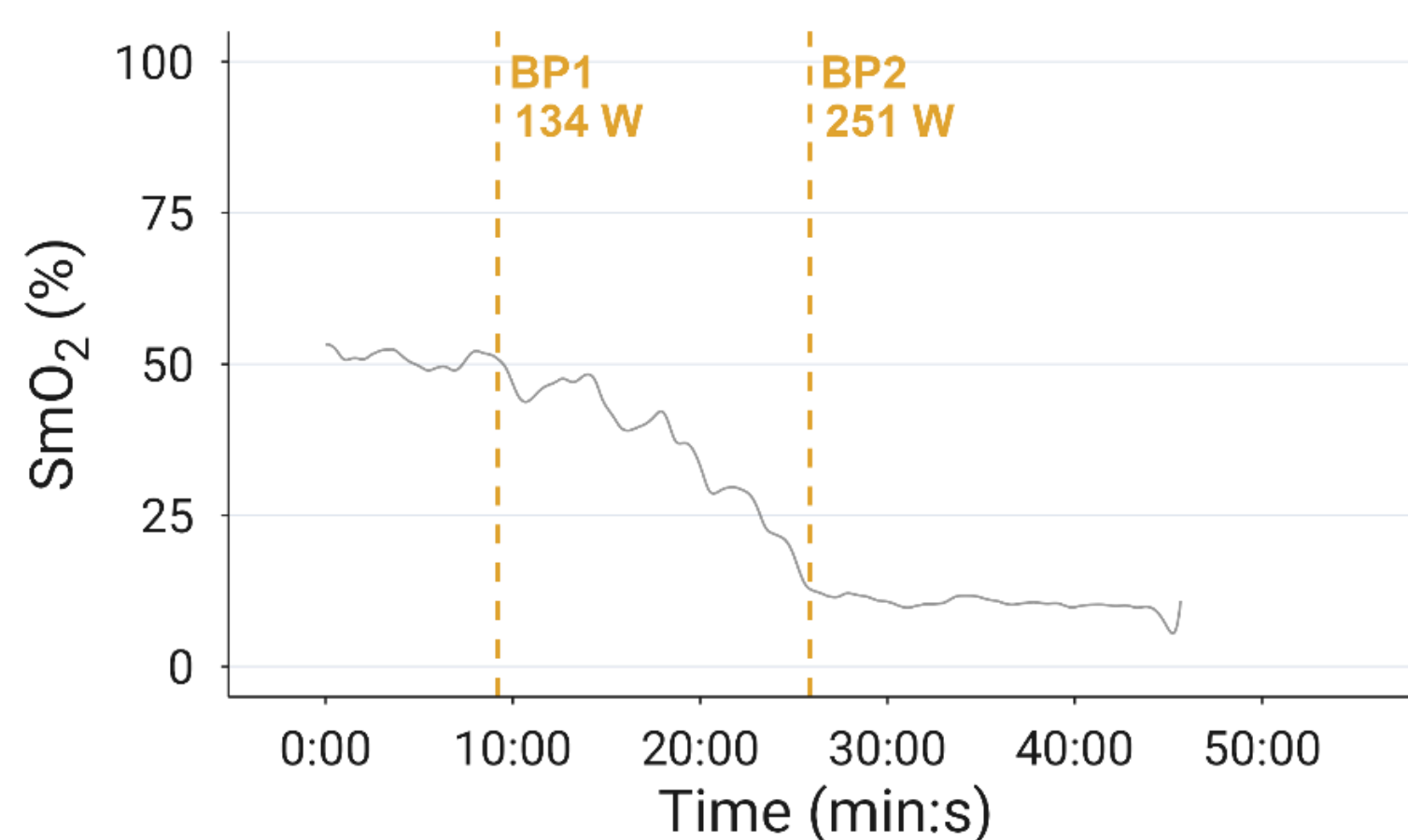
These BPs may offer muscle-specific insights into exercise tolerance beyond whole-body thresholds. Multiple algorithms exist to detect them,¹ with visual identification considered potentially the most physiological valid.²

3 – Aims & Methods

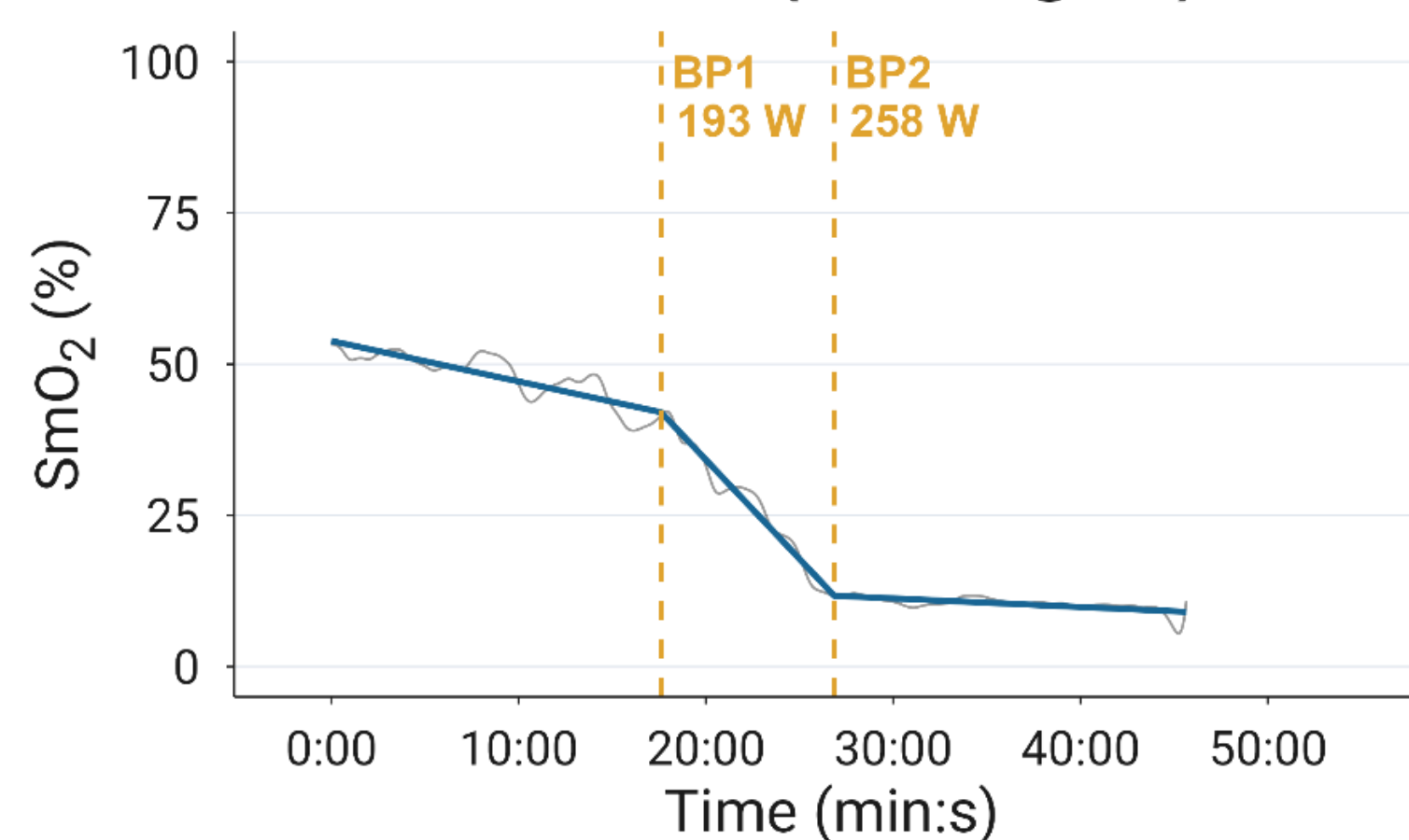
Aim: Compare test-retest reliability and identification success of SmO₂ BP detection methods; test for bilateral differences in BP power output.

Methods: Re-analysis of Skotzke et al. (2024)³ – n = 12 males, 3 GXTs each. VL SmO₂ BPs calculated for DOM, NDOM, AVG. Reliability: Coefficient of Variation (CV), SEM, ICC; Side/method: bootstrapped pairwise comparisons.

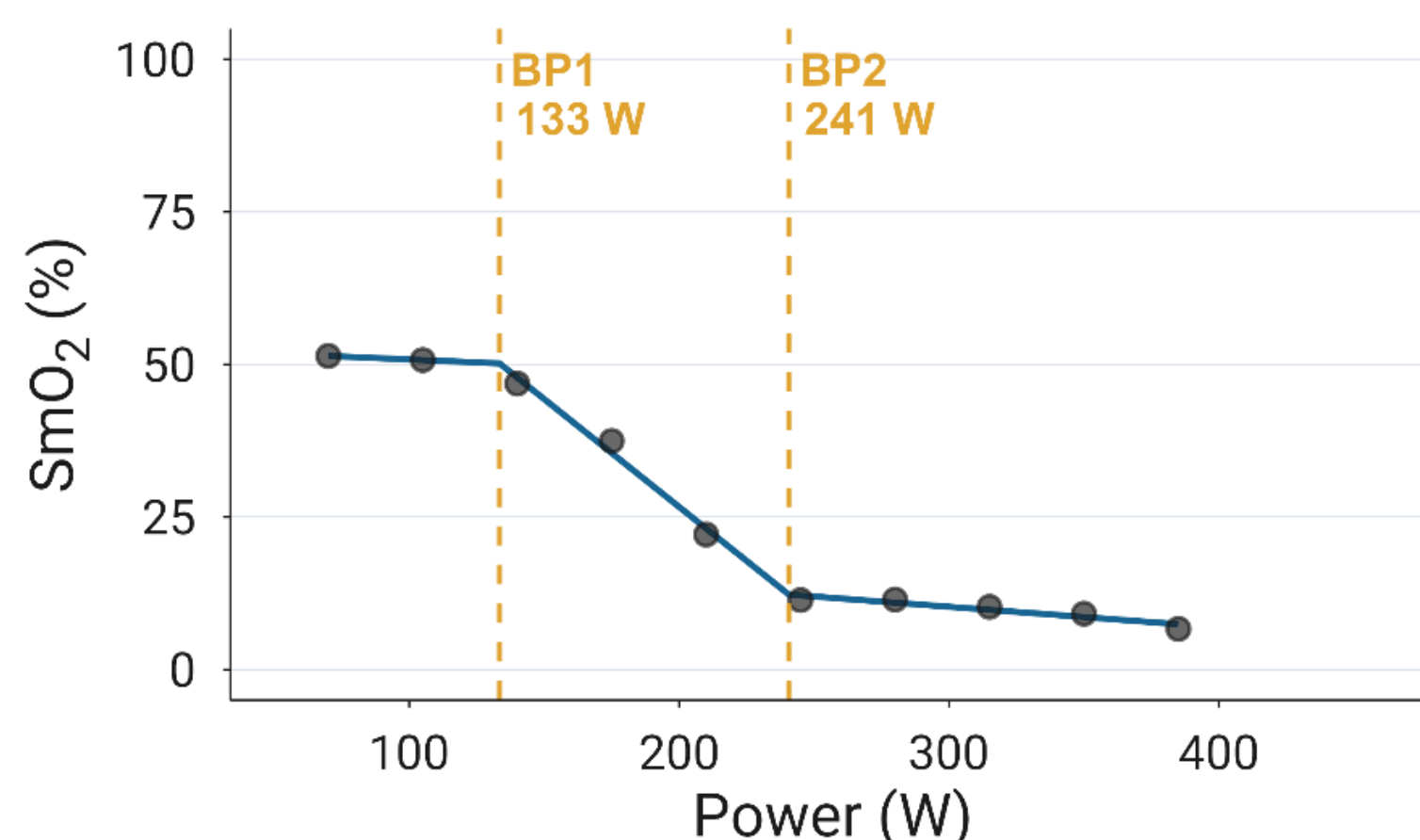
A - Visual identification



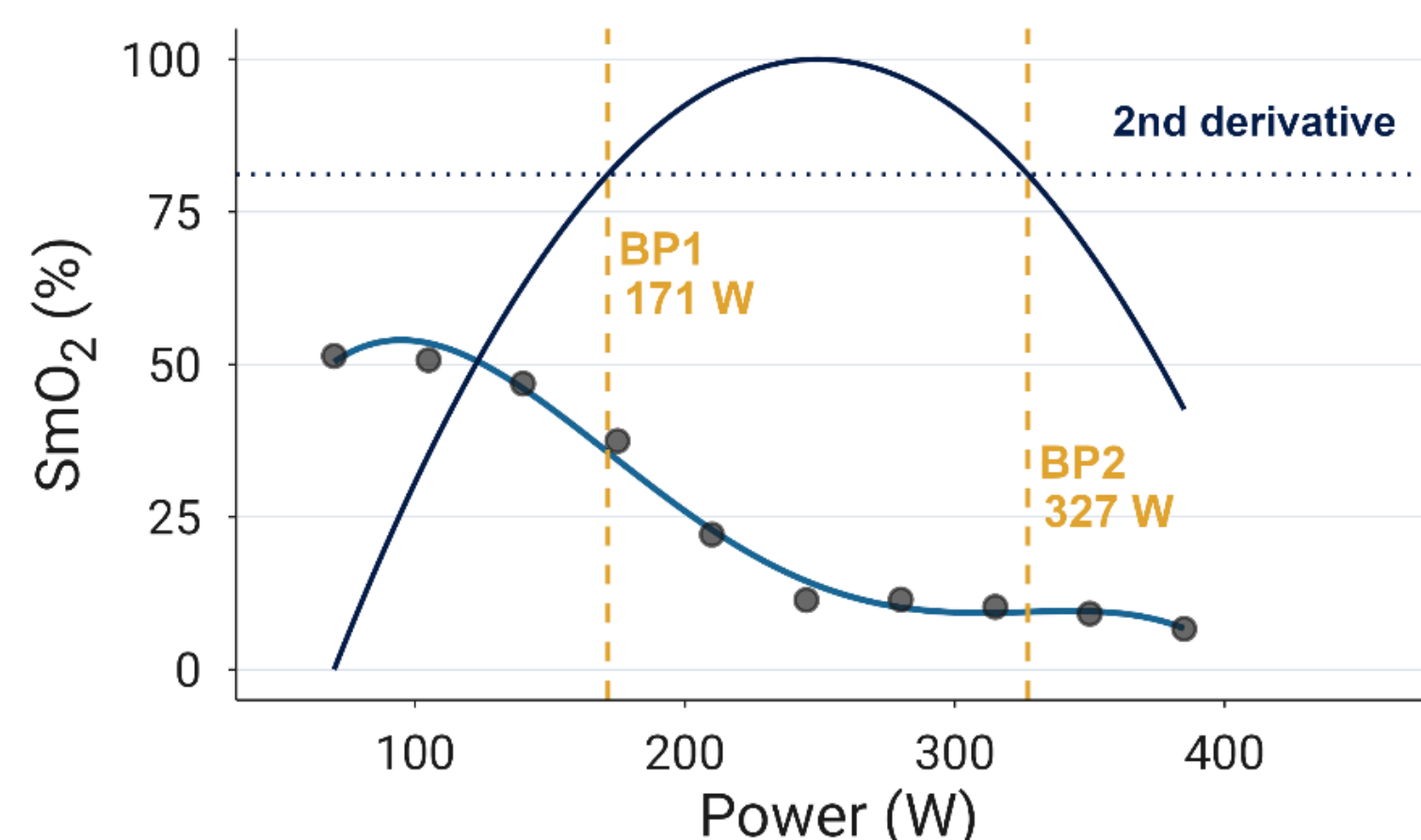
B - Double Piecewise (Raw signal)



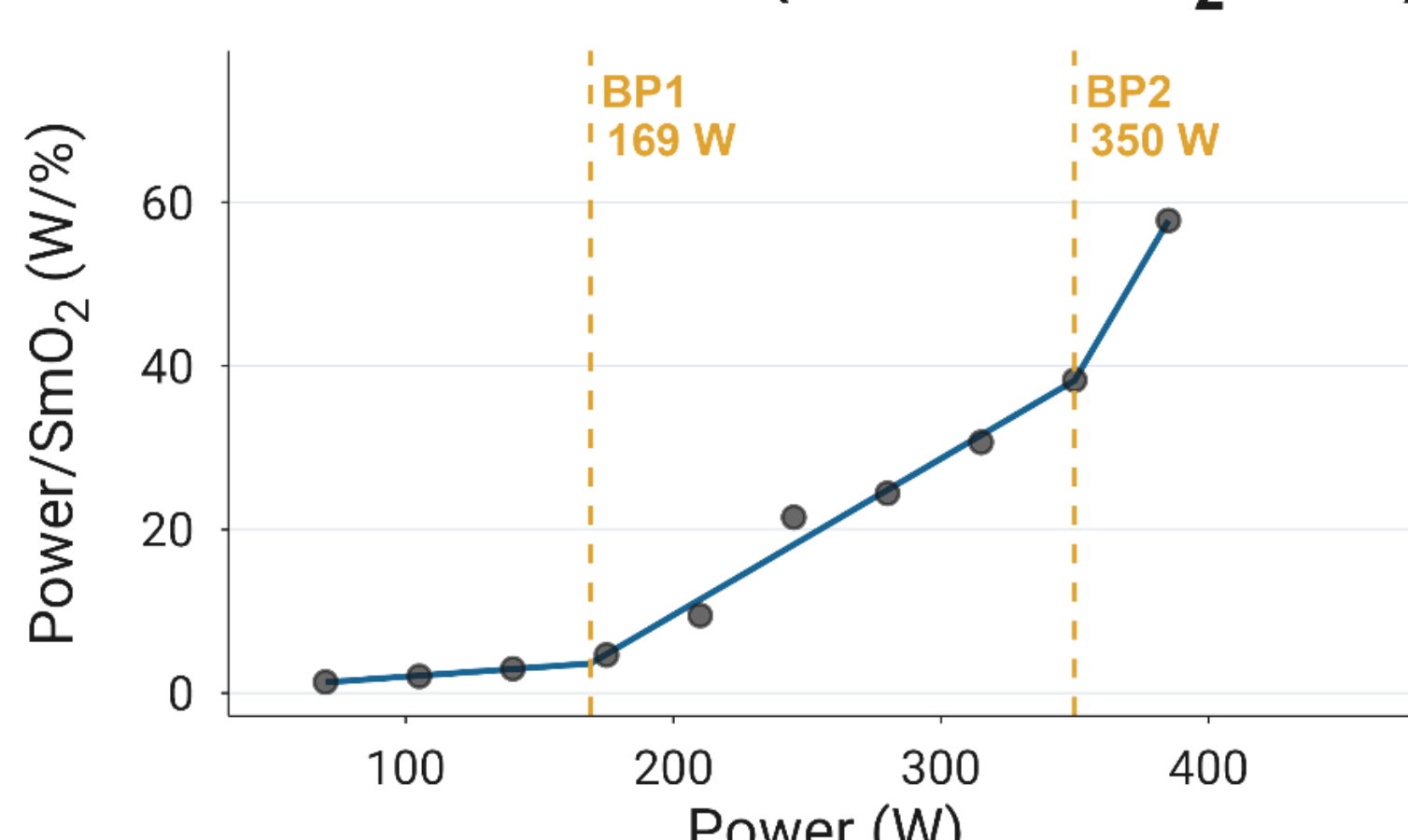
C - Double Piecewise (Stage means)



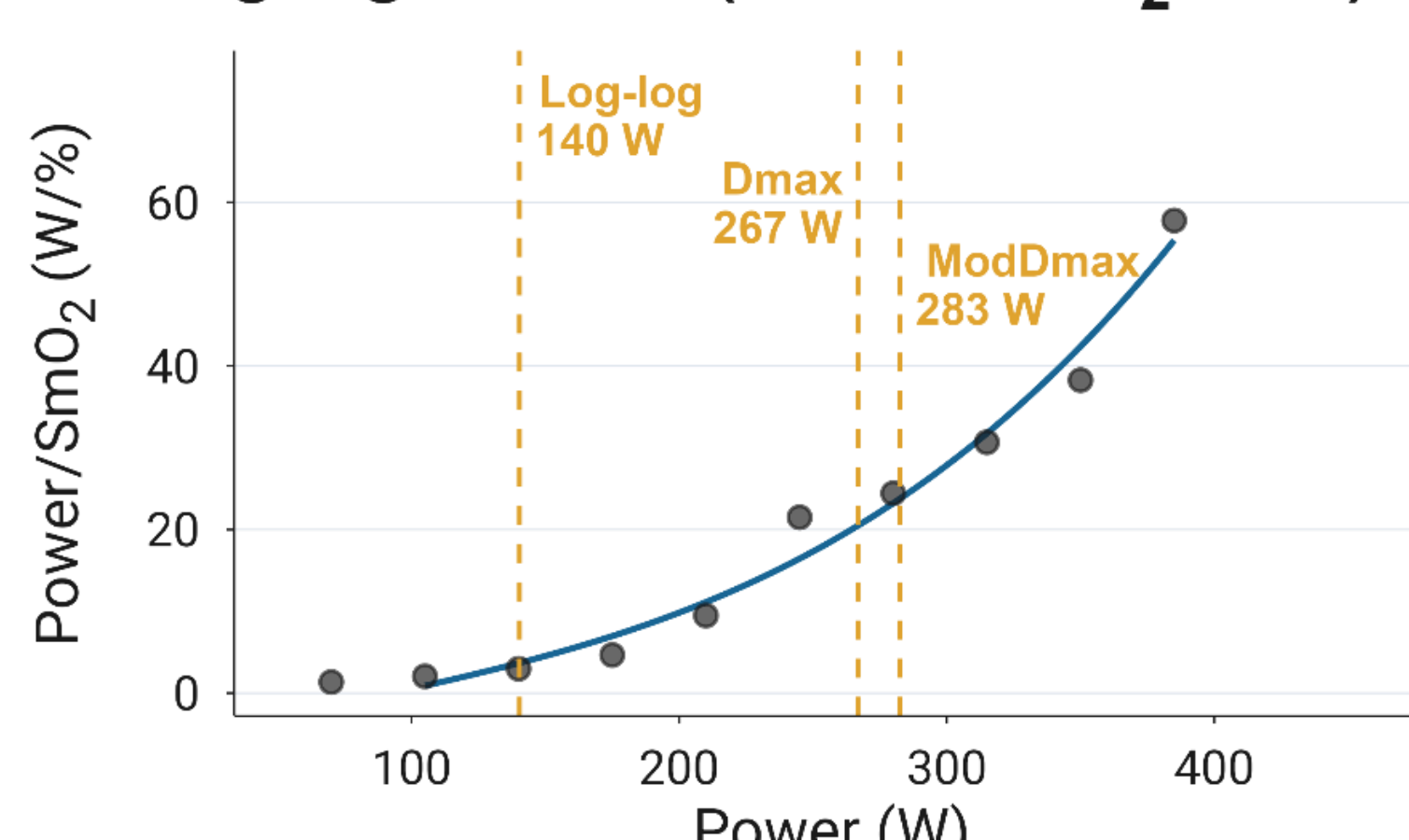
D - Polynomial inflections



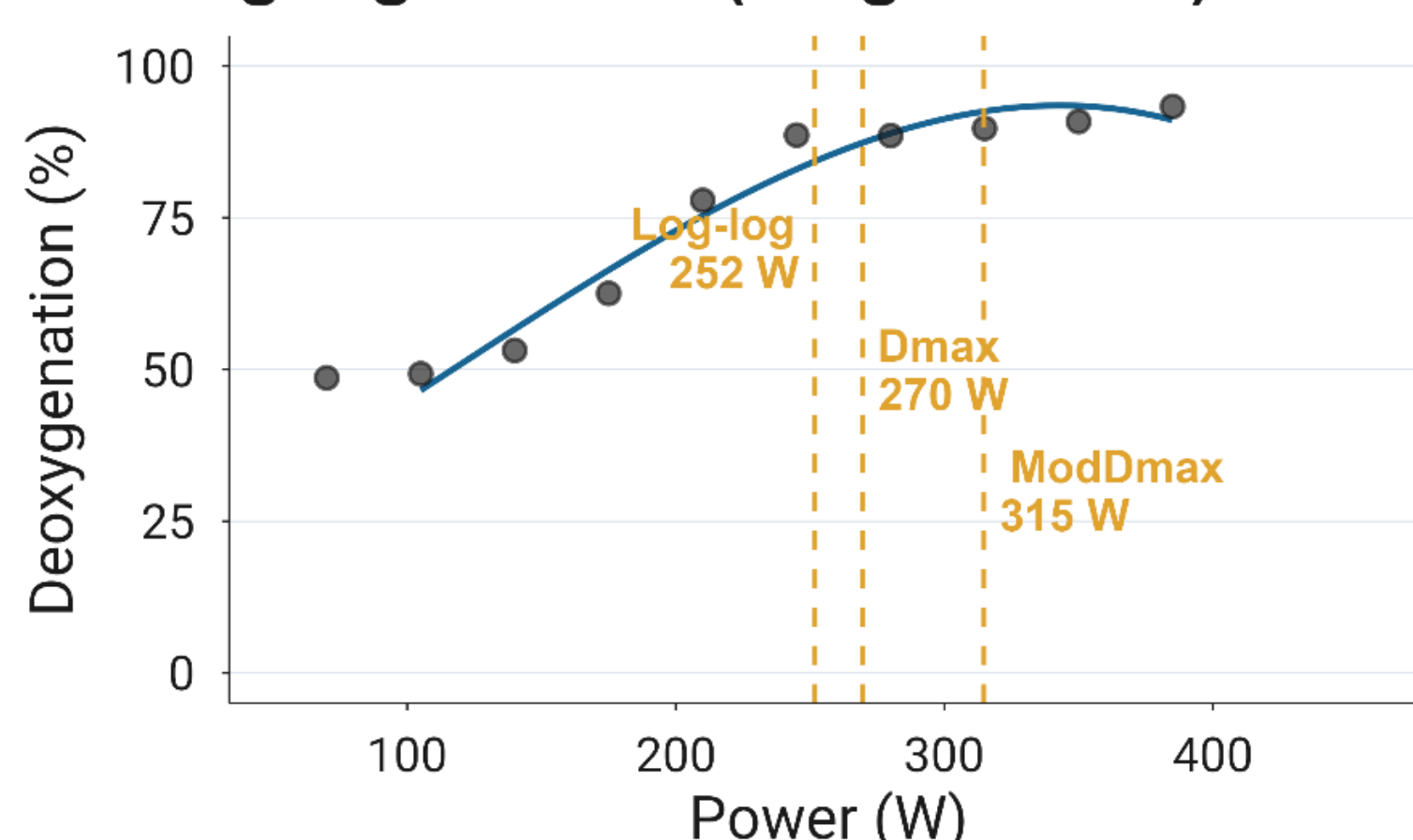
E - Double Piecewise (Power/SmO2 ratio)



F - Log-log & Dmax (Power/SmO2 ratio)



G - Log-log & Dmax (Stage means)



H - Double Piecewise (Slope)

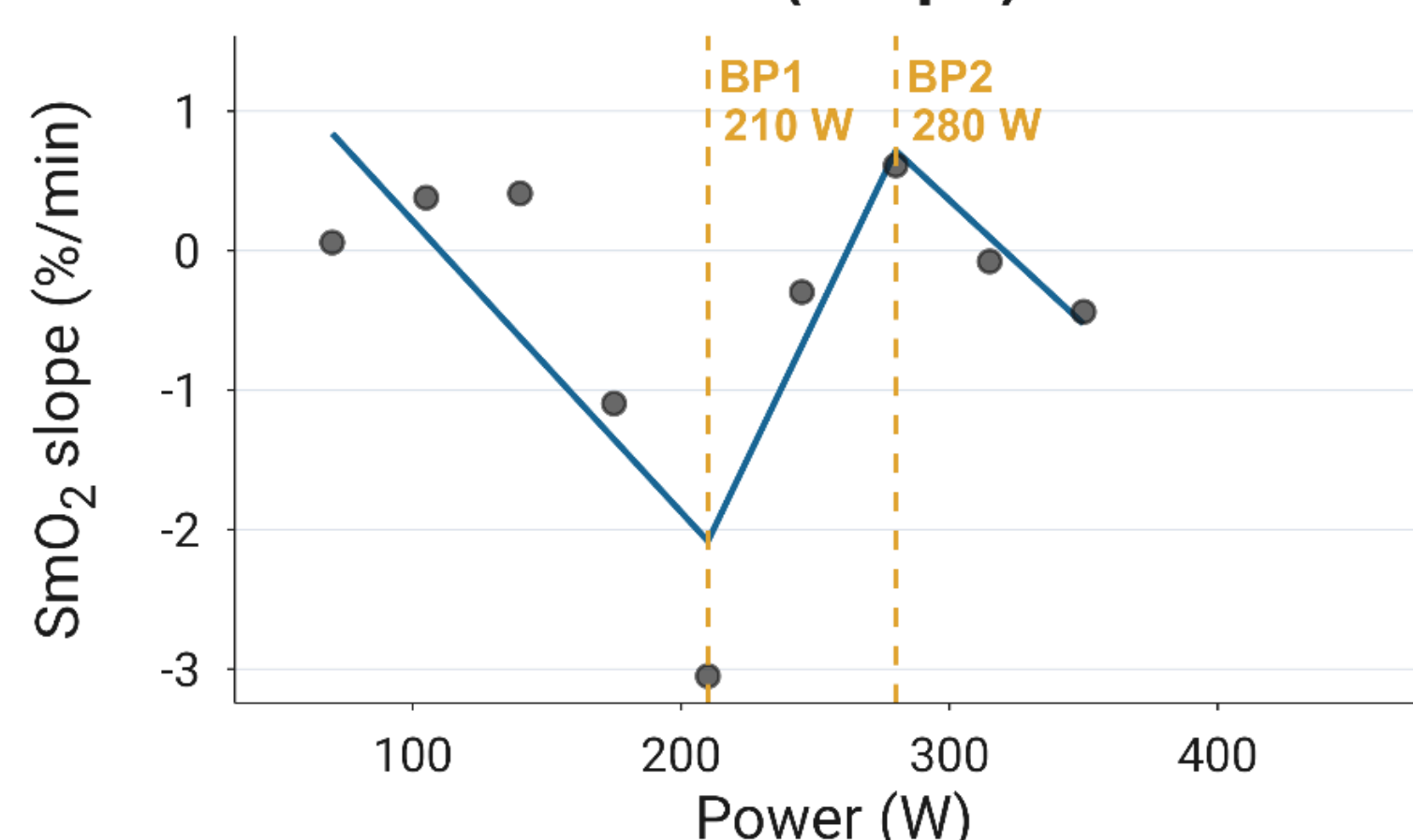


Fig. 1 Methods used for identifying SmO₂ breakpoint 1 (BP1) and 2 (BP2), shown for one DOM GXT. Dashed yellow lines indicate identified BPs

2 – Knowledge Gaps

The **test-retest reliability** of SmO₂ BP methods is largely unknown, a prerequisite for longitudinal application.

It is also unclear whether the BP power output differs between the dominant leg (DOM), non-dominant leg (NDOM), or their average (AVG).

4 – Results

• Slope & polynomial methods **failed to identify BPs in > 33 % of tests**

• ICC 0.25 – 0.88; CV 4.2 – 22.0 %; SEM 9 – 51 W

• Reliability differed significantly between methods in **17 – 19 % of pairwise comparisons**

• BP power output differed in **8 % of side comparisons**

A - BP1

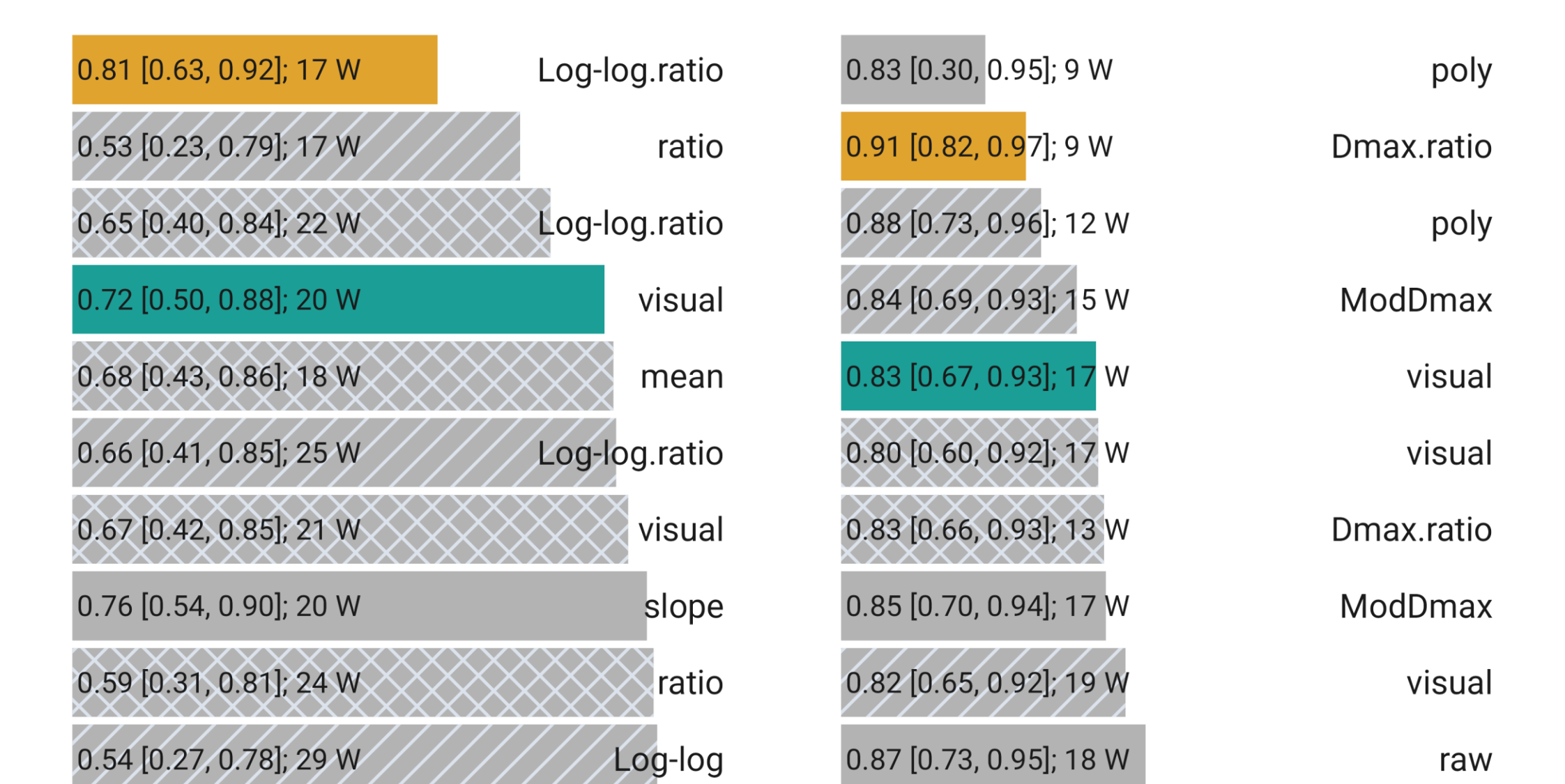


Fig. 2 10 lowest CV methods for BP1 (A) and BP2 (B); ICC [95% CI], SEM shown; lowest CVs: 8 % (BP1) and 4 % (BP2)

5 – Conclusions

• SmO₂ BP reliability and detection success **vary substantially by method**, which may explain inconsistent agreement with whole-body thresholds

• **Visual Identification** shows decent reliability and physiological validity

• Longitudinal analyses need to account for method-specific reliability

• No clear side differences – **measuring a single limb (VL) may be sufficient**

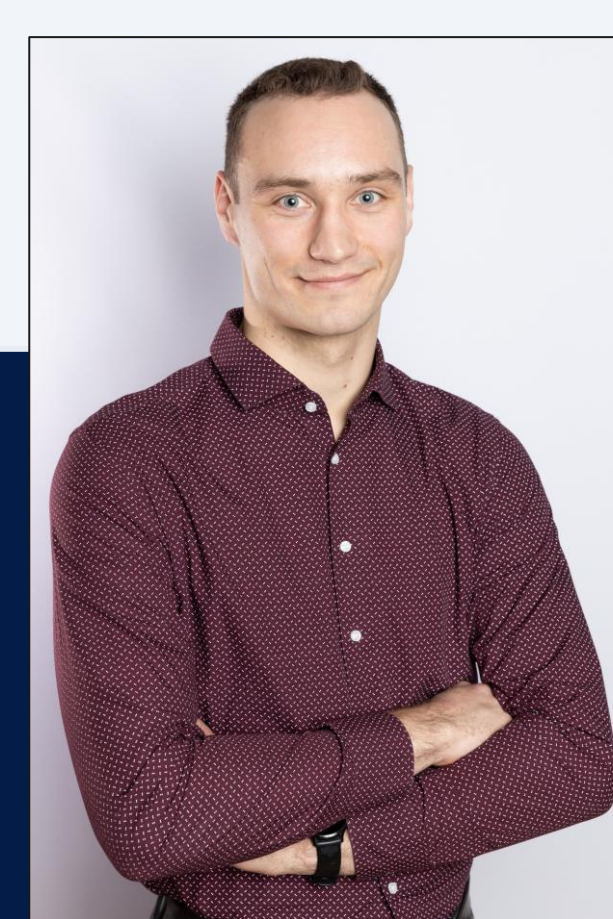
Choose a BP method based on physiological rationale, reliability, and concurrent validity.

References

- [1] Sendra-Pérez, C. et al. Sci Rep. 2023;13.
- [2] Fernández-Jarillo I., et al. Eur J Appl Physiol. 2026.
- [3] Skotzke P, et al. Eur J Appl Physiol. 2024;124(10).

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More Info on
Github

